Test Plan – Project 7

Stephen Wilton

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1. Test Plan Overview

This Test Plan outlines how testing will be conducted for the Project7 game by organizing existing Test Cases into execution scenarios. Rather than redefining individual tests, this plan focuses on how testers will execute them in a logical sequence that reflects real player behavior. These scenarios simulate typical gameplay, including navigation, combat, progression, and system interactions, to ensure the application functions correctly as a whole. Test results will be recorded in the Test Executions list to track system reliability and identify defects.

1. Application Description

A 2D wave-based survival game where the player fights enemies, upgrades stats, and progresses through waves.

Core features include:

- Player movement (WASD) and Dash (spacebar)
- Combat system (melee and ranged weapons)
- Enemy AI (basic, shooter, charger, boss)
- Wave progression
- HUD (health, XP, coins, timer)
- Shop and currency
- Level up and passive abilities
- Game over and restart system

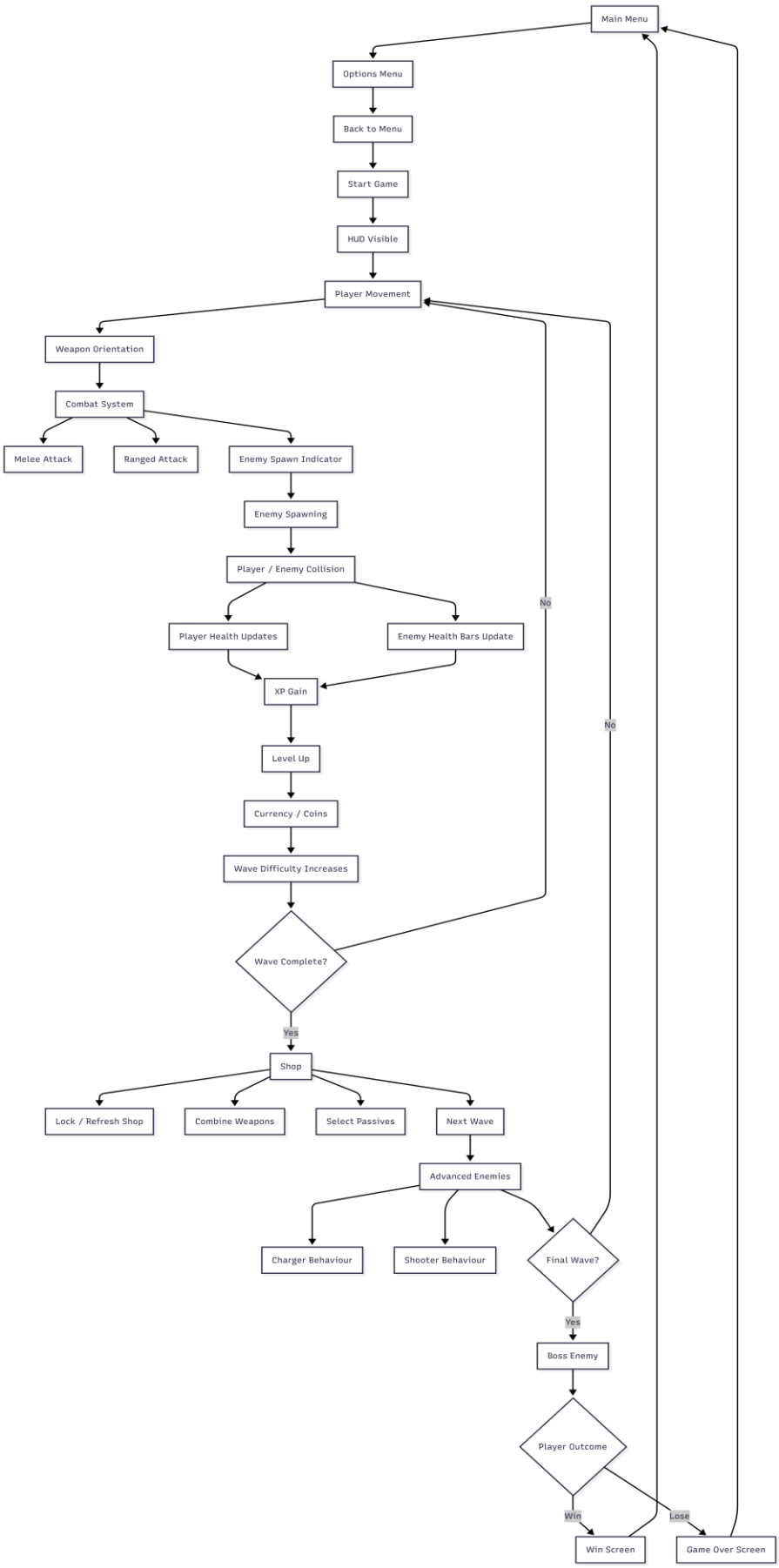
2. Test Environment

Component	Description
Platform:	Desktop (Windows)
Engine:	Godot
Database:	None
Input:	Keyboard (WASD, Spacebar, Esc, Enter) and Mouse

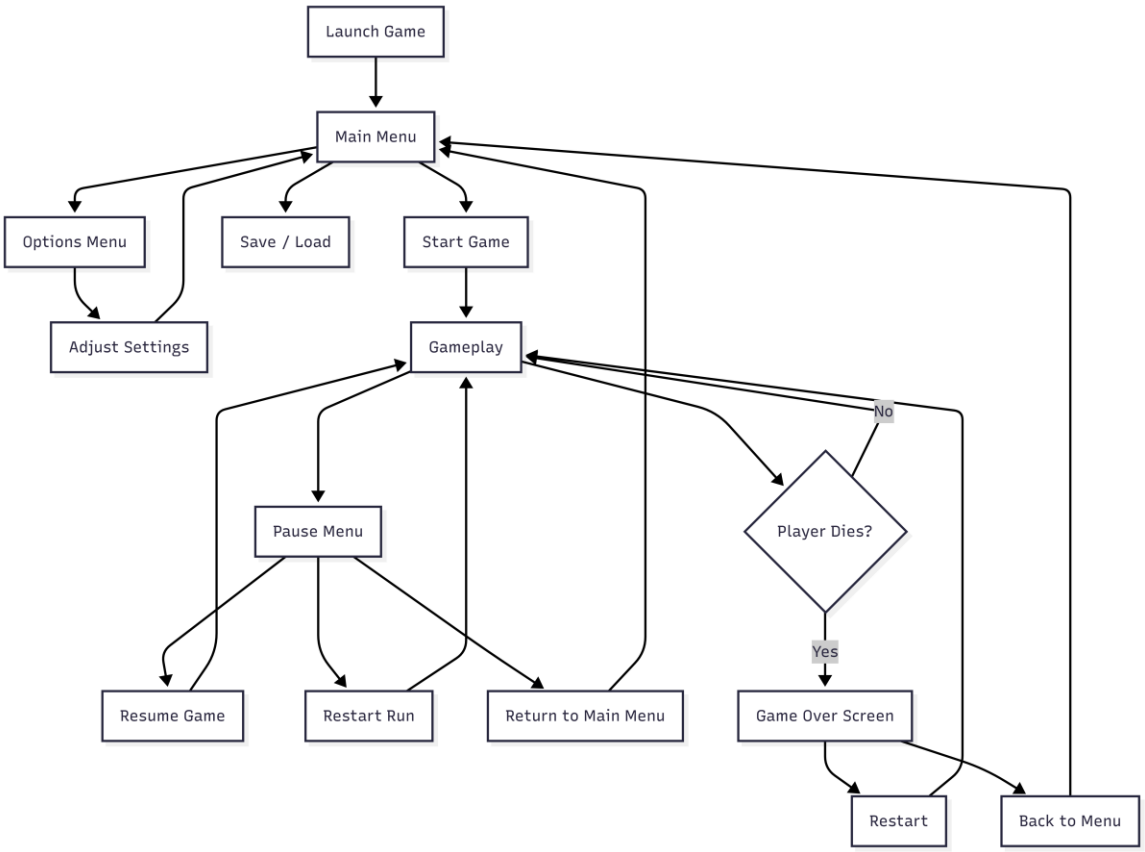
3. Execution Scenarios

The following diagrams represent the flow of events in certain areas of the game. Each node reflects a test case or is related to a test case.

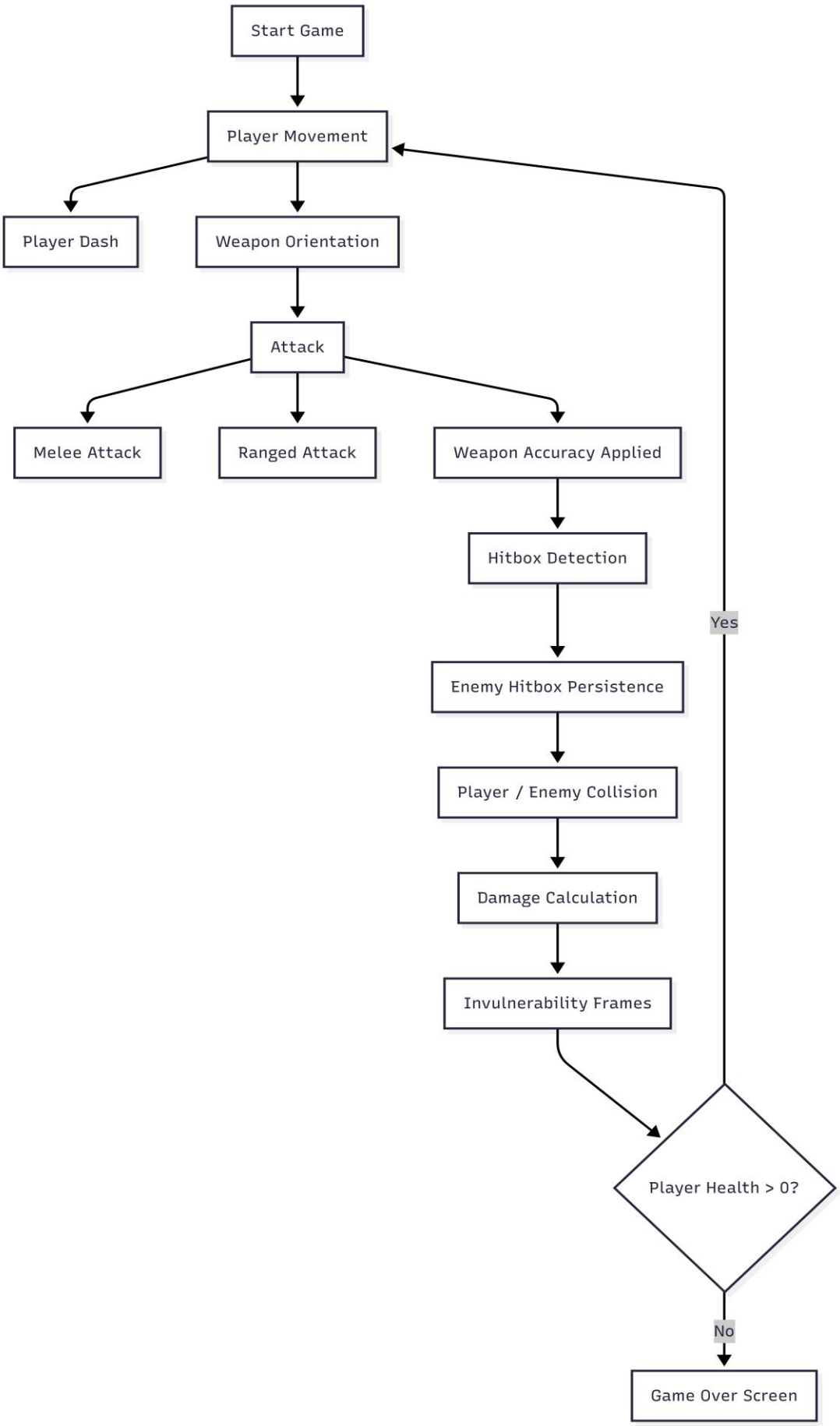
1. Main Gameplay Flow



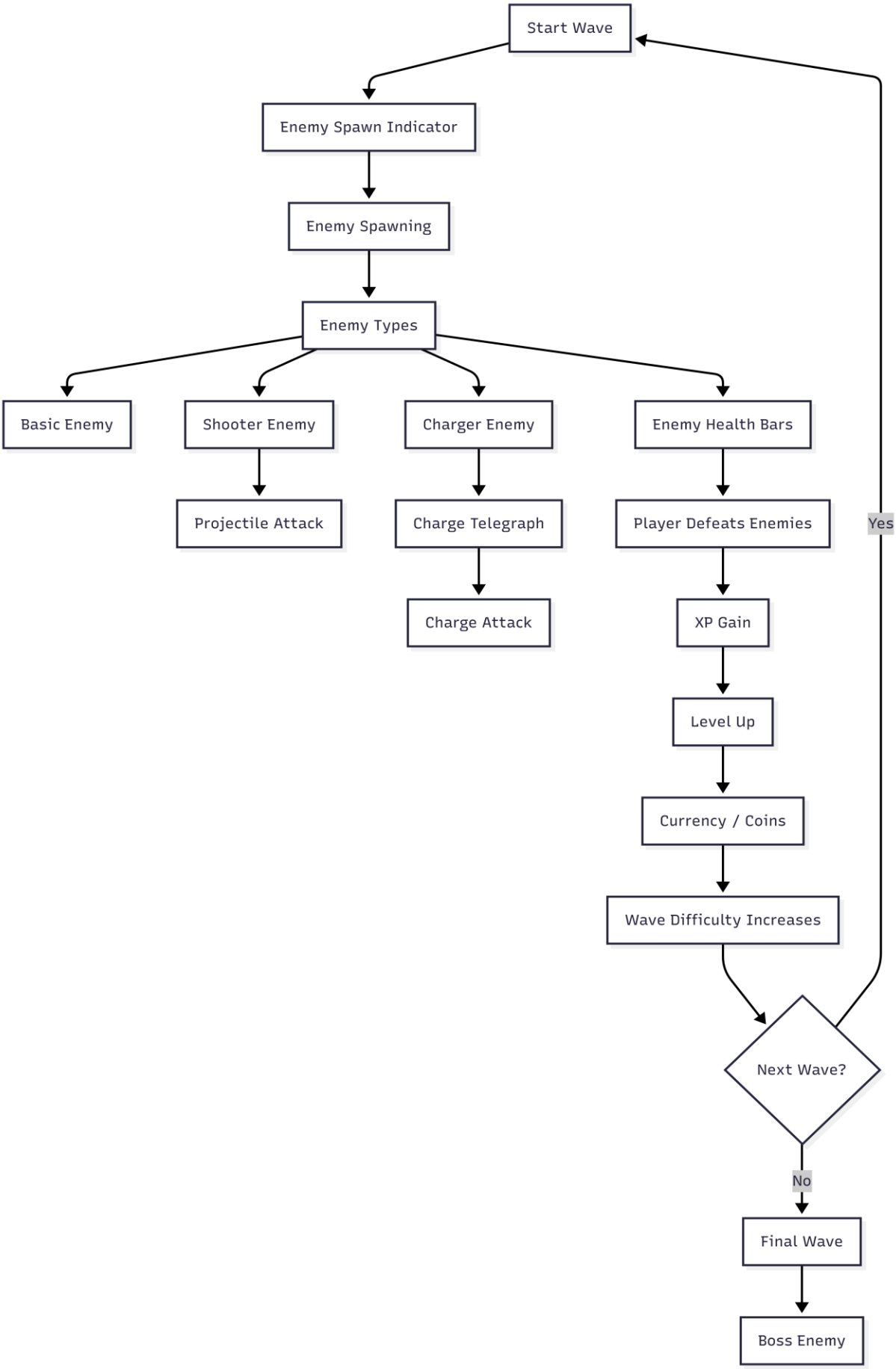
2. Menu / System navigation



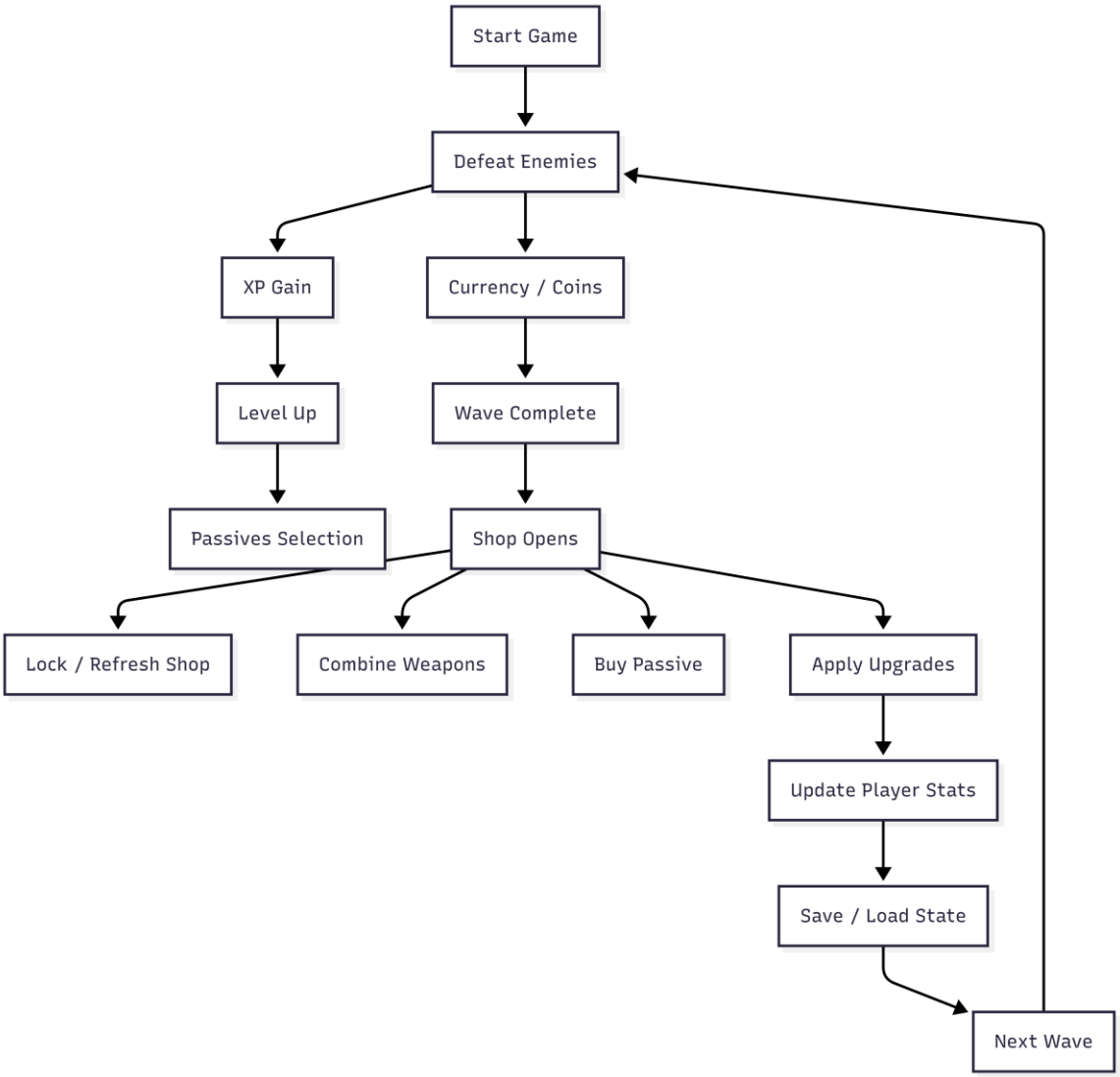
3. Player and Enemy Combat



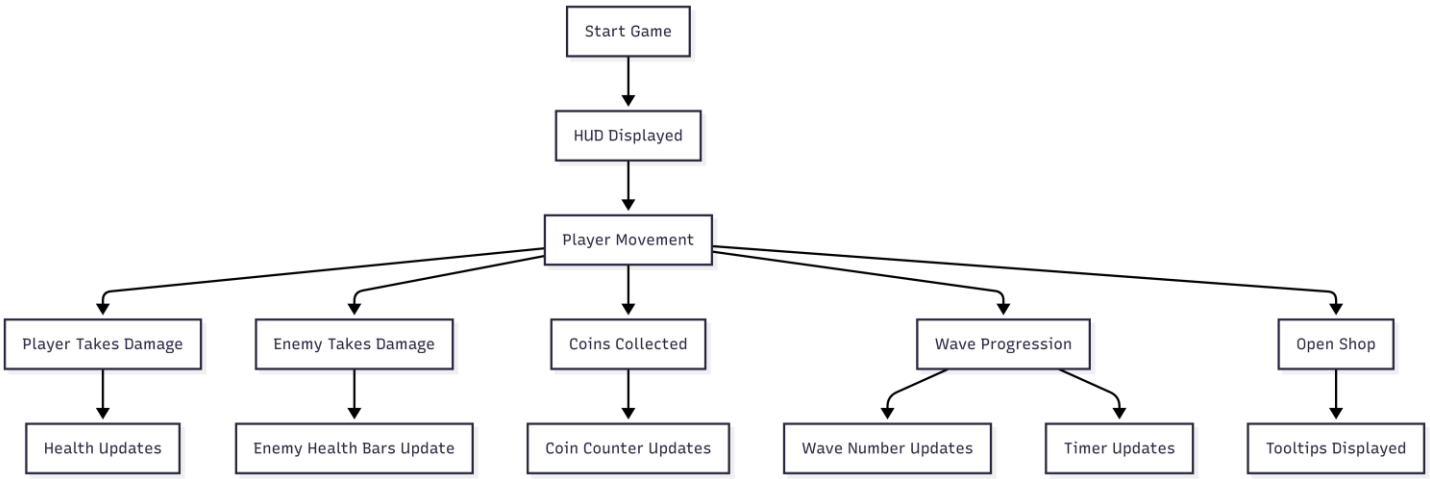
4. Enemy and Wave Behaviour



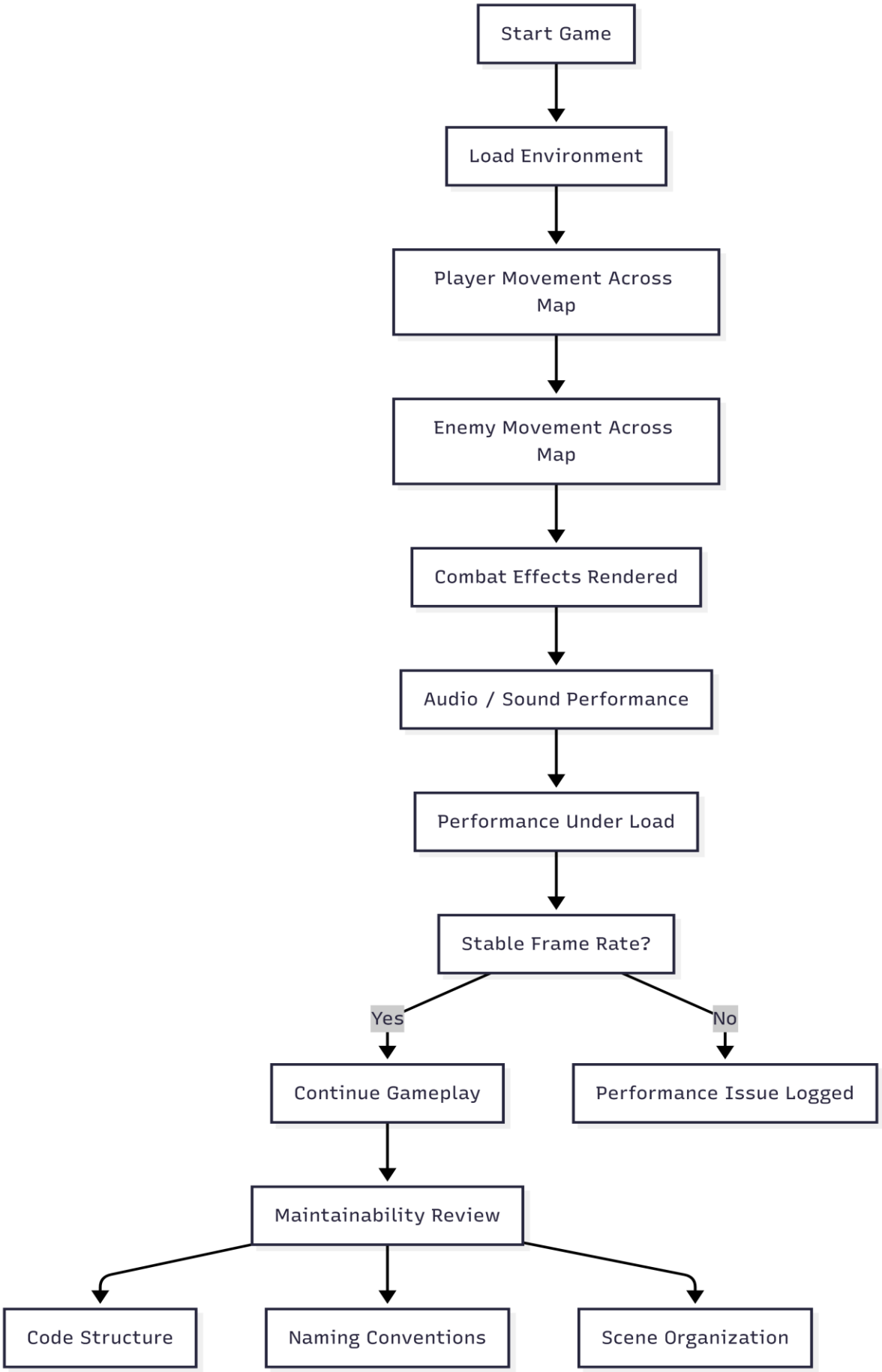
5. Player Progression and Shop



6. User Interface

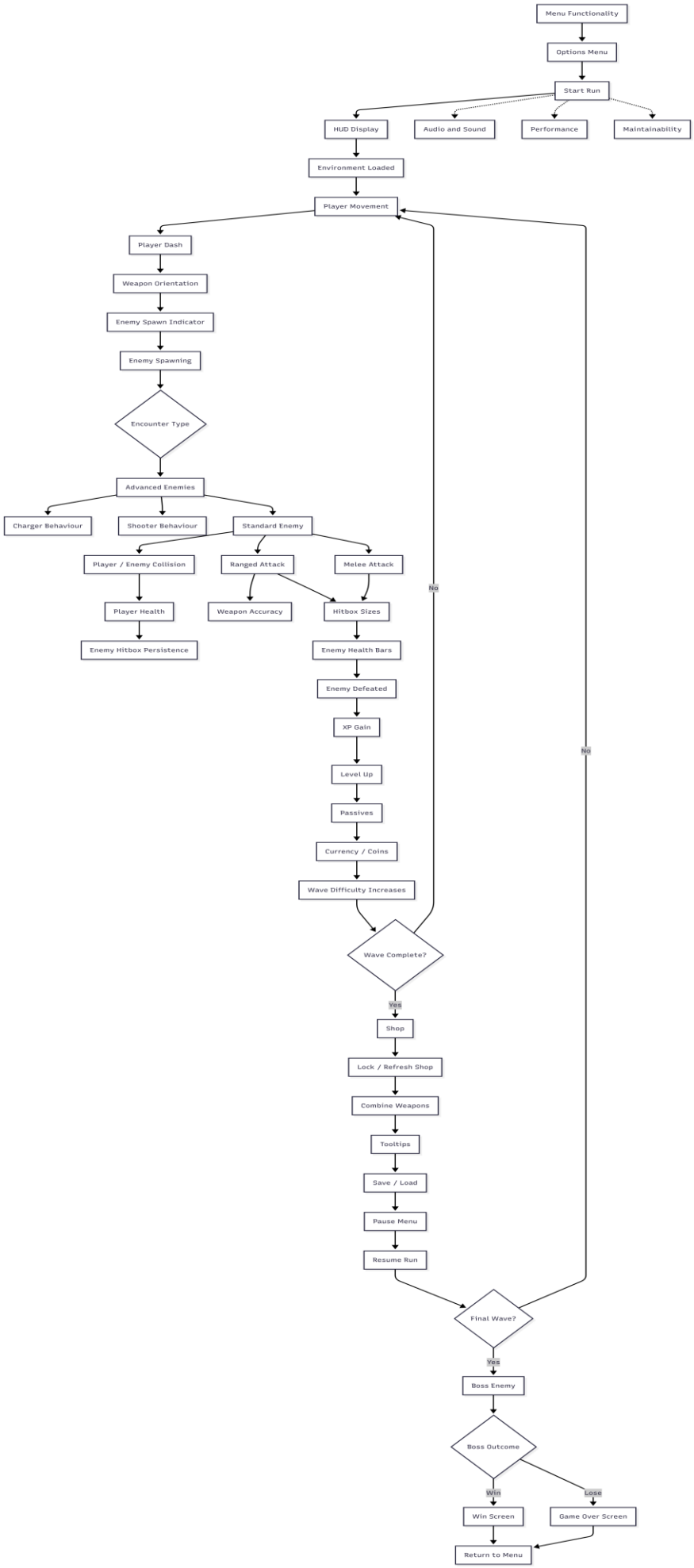


7. Environment and Non-functional Details



3.1 Complete Test Plan Scenario

The following is a complete flow of all events in the game. Each node is closely related to a requirement or a test case. You may need to zoom in to see all the details. If needed the image is available here on SharePoint:
CompleteDiagram.png


Menu Functionality
Options Menu
Start Run
HUD Display
Audio and Sound
Performance
Maintainability
Environment Loaded
Player Movement
Player Dash
Weapon Orientation
Enemy Spawn Indicator
Enemy Spawning
Encounter Type
Advanced Enemies
Charger Behaviour
Shooter Behaviour
Standard Enemy
Player / Enemy Collision
Ranged Attack
Melee Attack
Player Health
Weapon Accuracy
Hitbox Sizes
Enemy Hitbox Persistence
Enemy Health Bars
Enemy Defeated
XP Gain
Level Up
Passives
Currency / Coins
Wave Difficulty Increases
Wave Complete?
No
Yes
Shop
Lock / Refresh Shop
Combine Weapons
Tooltips
Save / Load
Pause Menu
Resume Run
Final Wave?
No
Yes
Boss Enemy
Boss Outcome
Win
Lose
Win Screen
Game Over Screen
Return to Menu

4. Associated Test Cases

Each of the above Execution Scenarios has the following test cases associated with it:

Menu / System Navigation

- Menu Functionality
- Options Menu
- Save / Load
- Pause Menu
- Game Over Screen

Player Combat and Survivability

- Player Movement
- Player Dash
- Weapon Orientation
- Melee Weapon
- Ranged Attack Extra Stats
- Weapon Accuracy
- HitBox Sizes
- Player / Enemy Collision
- Player Health
- Enemy Hitbox Persistence

Enemy and Wave Behaviour

- Enemy Spawning Indicator
- Advanced Enemies
- Charger Behaviour
- Enemy Shooter Behaviour
- Enemy Health Bars
- Wave Difficulty Increases
- Boss Enemy

Progression / Shop Flow

- Experience and Levelups
- Currency and Coins
- Passives
- Shop Lock and Refresh
- Combine Weapons

UI / Feedback

- Heads Up Display
- Tooltips
- Audio and Sound

Environment / Non-Functional

- Environment
- Performance
- Maintainability

7. Test Execution Process

1. Locate the Test Case in SharePoint
2. Execute the steps
3. Record result: Pass / Fail / Blocked
4. If failed, record defect details

8. Completion Criteria

- All Test Cases are executed
- No critical defects remain
- Core gameplay loop is stable